

1(a)	any one from: <u>mitochondrion</u> ; bod <u>mitochondria</u> chloroplast ; nucleus ; R newcleus / nuculus / nucelus	1
1(b)(i)	structure A = capsid ; A protein coat I other correct detail e.g. (<i>capsid</i>) <i>made of, protein / subunits / capsomeres</i> or (<i>capsid / outer coat</i>) <i>with capsomeres / containing capsomeres</i> R capsid with, envelope / wall / outer membrane structure B = (murein / peptidoglycan / bacterial) cell wall ; I outer layer R cellulose / chitin, cell wall bod murein / peptidoglycan, wall	2
1(b)(ii)	152 – 186 (nm) to a <u>whole number</u> ;; <i>one calculated value required</i> (X–Y 63–65 mm and <i>T-phage image length 11–13 mm</i>) <i>allow 1 mark if correct but not to nearest whole number</i> or <i>allow 1 mark if answer is a whole number between 136–151 or 187–203 and X–Y measurement recorded as 62 or 66 mm and/or T-phage length 10 or 14 mm</i>	2

Question	Answer	Marks
1(c)	<i>look for ora if describing why T-phage does not bind to other species</i> 1 <i>ref. to specific (to <i>E. coli</i>) / specificity (of T-phage to <i>E. coli</i>) ;</i> A <i>ref. to complementarity / complementary (shape for binding)</i> 2 <i>detail of T-phage <u>and</u> <i>E. coli</i> interaction ;</i> R <i>if describing enzyme-substrate <u>or</u> antibody-antigen binding</i> <i>T-phage (ends of) tail fibres / (T-) phage / receptors (on phage)</i> I <i>antigen R membrane</i> A <i>'it' if correct context (T-phage is in the Q)</i> <i>named location on <i>E. coli</i> (to which the phage attaches)</i> <i>receptor (in / on, cell wall / cell surface membrane)</i> A <i>cell receptor but R receptor cells</i> <i>cell (surface) membrane</i> <i>outer membrane / lipopolysaccharide / porin (of cell wall)</i> <i>membrane protein / glycoprotein</i> bod <i>antigen <u>context is antigenic molecule on the surface</u></i> I <i>cell wall / murein / peptidoglycan</i>	2
1(d)(i)	<i>malaria <u>and</u></i> <i>not caused by, a bacterium / bacteria / prokaryote</i> or <i>caused by a, protoctist / eukaryote ;</i> R <i>if 'virus' incorrectly appears within answer</i> I <i>details about malaria e.g. antigenic concealment / vector</i> bod <i>close or phonetic spelling for protoctist and eukaryote</i> <i>allow non-syllabus names e.g. protozoa / protista / apicomplexa / sporozoa</i>	1

Question	Answer	Marks
1(d)(ii)	accept AW and ora for these mps any three from: 1 will not harm / AW, human / body, cells ; A tissues / organs, for cells 2 less / no, chance of, allergic reactions / immune response / side effects / toxic effects / AW (from treatment) ; 3 specific to treat a particular disease / kills only the pathogen intended ; I specific to bacteria <i>context is specific for / to kill only, the bacteria causing the disease</i> 4 does not, affect / kill, beneficial / AW, bacteria ; A gut microbiome not affected 5 can, replicate / multiply, within person being treated (as they replicate in cells of bacterial pathogen) ; I reproduces 6 (may) need only one dose / short(er) duration of treatment ; A idea of no need for a, full / long, course (compared with antibiotics) 7 (may result in) quicker cure / faster recovery / faster response <i>(than, antibiotics / passive immunisation / antibody treatment) ;</i> 8 can use to treat antibiotic-resistant bacteria / decreases impact of antibiotic resistance / decreases chance of antibiotic resistance occurring / AW ; 9 detail ; e.g. fewer antibiotics prescribed (<i>not having a long course is mp6</i>) less reliance on antibiotics eases pressure on, finding / developing, new antibiotics 10 less / no, risk of becoming resistant to attack by T-phage ; 11 AVP ; e.g. less research and development required	3